

Lab 31: Saving and Loading Data with NumPy

Aim

To save NumPy arrays in **binary and text formats** and reload them for later use.

Algorithm

1. Import the NumPy library.
2. Create a NumPy array.
3. Save the array in binary .npy format using np.save().
4. Save the array in CSV format using np.savetxt().
5. Load the .npy file using np.load().
6. Load the CSV file using np.loadtxt().
7. Display the loaded arrays.

Program

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```
import numpy as np
```

```
# Create a NumPy array
```

```
a = np.array([  
    [1, 2, 3],  
    [4, 5, 6]  
])
```

```
# Save array in binary format
```

```
np.save("array.npy", a)
```

```
# Save array in CSV format
```

```
np.savetxt("array.csv", a, delimiter=",")
```

```
# Load saved arrays
```

```
loaded_numpy = np.load("array.npy")
```

```
loaded_csv = np.loadtxt(
```

```
    "array.csv",
```

```
    delimiter=","
```

```
)
```

```
# Display loaded arrays
```

```
print("Loaded from .npy:\n", loaded_numpy)
```

```
print("\nLoaded from .csv:\n", loaded_csv)
```

Sample Output

Loaded from .npy:

```
[[1 2 3]
```

```
[4 5 6]]
```

Loaded from .csv:

```
[[1. 2. 3.]
```

```
[4. 5. 6.]]
```

Result

Thus, NumPy arrays were successfully saved and loaded using both binary (.npy) and text (.csv) file formats.